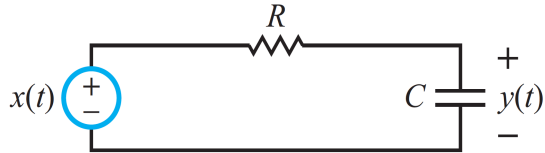


1. (40 pts) For the following RC circuit, calculate the following outputs for response $x(t) = (1 + \sin^2(2t))u(t)$, when $R=250\text{k}\Omega$, $C=1\mu\text{F}$, and an initial capacitor voltage of 1V.



- Zero-input response
 - Zero-state response
 - Transient response
 - Steady-state response
 - Natural response
 - Forced response
2. (40 pts) Given an LTI system with an impulse response $h(t) = 2\delta(t) - e^{-2t} \cos(2t) u(t)$, determine the following:
- $\mathbf{H(s)}$
 - $\hat{\mathbf{H}}(\omega)$
 - Poles and zeros of the system
 - LCCDE description of the system
 - Output response $y(t)$ to input $x(t) = 2(t - 1)e^{-3t}u(t)$. Assume zero initial conditions.
3. (20 pts) Rewrite the following circuit in the $\mathbf{s}$ -domain.
Initial conditions: Capacitor voltage = 0V. Inductor current = -0.1A

